Python Activity-Based Learning
49. Tuple-Based Product Catalogue

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement
A shop stores product details as tuples. Each tuple contains a product name, price, and stock quantity. Write a program to calculate the total stock value and show products that need restocking.

2. Objectives
• Store product details in tuples.
• Calculate the value of each product's stock.
• Find products with stock below five units.

3. Assumptions
• Each product is stored as (name, price, quantity).
• A product needs restocking when its quantity is less than 5.
• Prices and quantities are not negative.

4. Algorithm
1. Start.
2. Create a list of product tuples.
3. Set total stock value to zero.
4. Read each tuple using tuple unpacking.
5. Multiply price by quantity and add it to the total.
6. If quantity is less than five, display the product name.
7. Display the total stock value.
8. Stop.

5. Flowchart
START → Read product tuples → Unpack name, price, quantity → Calculate price × quantity → Is quantity below 5? → Yes: show restock message / No: continue → Display total stock value → STOP

6. Pseudocode
```
START\nproducts = list of (name, price, quantity) tuples\ntotal_value = 0\nFOR each tuple in products\n    UNPACK name,
```

7. Python Source Code
```
products = [
    ("Pen", 10, 20),
    ("Notebook", 50, 3),
    ("Pencil", 5, 4)
]

total_value = 0

for name, price, quantity in products:
    value = price * quantity
    total_value += value
    print(name, "stock value:", value)

    if quantity < 5:
        print(name, "needs restocking")

print("Total stock value:", total_value)
```

8. Test Cases

Input / Test	Expected result	What it checks
Pen: 10 × 20	200	No restock warning
Notebook: 50 × 3	150	Restock warning
All three products	Total value 370	Combined stock value

9. Sample Output
```
Pen stock value: 200\nNotebook stock value: 150\nNotebook needs restocking\nPencil stock value: 20\nPencil needs restock
```

10. Conclusion
This program uses tuples, tuple unpacking, lists, and conditions to check product stock and calculate stock value.